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ADIABATIC PRE-COOLING FOR A GAS COOLER

Abstract

A cooling device may include a water header pipe with drain holes arranged in a row oriented at an angle. The water header pipe may receive a supply of water and discharge a first portion of water through the drain holes. An evaporative cooling medium may include a first media portion and a second media portion. The first media portion may have a first media body to receive the first portion of water and discharge a third portion of water. The second media portion may have a second media body to receive the third portion of water and conduct a fourth portion of water. The second media body may receive a first inflow of air at a first temperature and discharge a first outflow of air at a second temperature that is lower than the first temperature by evaporative cooling of the fourth portion of water.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to cooling systems and more particularly to improved adiabatic pre-cooling for a gas cooler.

BACKGROUND

[0002] Gas cooling systems are used to reduce the temperature of a circulating coolant in gas form, typically without changing the state of the coolant from gas to liquid. Circulating coolant in hot gas form, such as carbon dioxide or other materials and compounds, may be heated in various ways such as by a compressor, by passing through a heat exchanger, by absorbing heat from warmer region such as an interior space of a home or office building, by other heat generating or heat transferring processes and systems, or a combination of these. A hot gas may pass through a pipe with attached fins so that heat from the hot gas may conduct to the attached fins and be transferred to a lower temperature environment by convection and/or radiation so that the temperature of the hot gas is reduced.

[0003] Adiabatic cooling has been used to provide additional cooling in various applications, where a water header pipe may discharge water onto a portion of an evaporative cooling pad by discharging the water in a directly downward direction in order to maximize drainage from the water header pipe, for example.

SUMMARY

[0004] To improve gas cooling, this disclosure provides various cooling systems, cooling devices, and cooling methods that utilize adiabatic cooling.

[0005] According to an example, a cooling device may include a water header pipe having a plurality of drain holes arranged in a row that may be parallel with a central axis of the water header pipe, the row may be oriented at an angle measured between a radius to the central axis of the water header pipe and a vertical plane passing through the central axis, the water header pipe may be configured to receive a supply of water and discharge a first portion of water through the plurality of drain holes; and an evaporative cooling medium may comprise a first media portion and a second media portion, the first media portion may have a first media first end, a first media second end, and a first media body, the first media first end may be configured to receive the first portion of water, the first media body may be configured to conduct the received first portion of water within the first media body as a second portion of water, the first media second end may be configured to discharge a third portion of water; and the second media portion may have a second media first end, a second media second end, and a second media body, the second media first end may be configured to receive the third portion of water, the second media body may be configured to conduct the received third portion of water within the second media body as a fourth portion of water, wherein the second media body may be configured to receive a first inflow of air at a first temperature and discharge a first outflow of air at a second temperature that is lower than the first temperature, the first outflow of air may be cooled by evaporative cooling of the fourth portion of water.

[0006] According to this example, the cooling device wherein the angle may be between at least 10-degrees to not more than 40-degrees. The cooling device wherein the first portion of water may be directed to a first media portion exterior edge, and wherein the second portion of water may be configured to initially saturate a second media portion exterior surface. The cooling device wherein the first media body may include a first plurality of channels arranged in a substantially vertical manner, and wherein the second media body may include a second plurality of channels arranged in a substantially horizontal manner, orthogonal to the first plurality of channels. The cooling device wherein at least one of the first media portion and the second media portion may include cellulose. The cooling device may further include an elongated deflector shield disposed to at least partially surround an upper portion of the water header pipe, the elongated deflector shield may be configured to at least one of protect the upper portion of the water header pipe and redirect water from the water header pipe in a direction toward the first media first end.

[0007] According to this example, the cooling device wherein the elongated deflector shield may include a lower portion with a lower edge portion that may be laterally displaced from a second vertical plane that may be coplanar with an outer surface of the first media portion. The cooling device wherein the second media second end may be configured to discharge a fifth portion of water, the device may further include a reservoir configured to collect the fifth portion of water as a collected fifth portion of water; and a pump configured to transport the collected fifth portion of water and provide the supply of water to the water header pipe. The cooling device wherein the pump may have a pump profile that is limited to at least one of avoid overspray, avoid bouncing off (e.g., ricochet) first media first end, and promote capture and conduction of water in the first media portion. The cooling device wherein the supply of water may be provided to one of a center portion of the water header pipe and an end portion of the water header pipe, wherein at least one end portion of the water header pipe may be closed.

[0008] According to this example, a cooling system may include the cooling device and may further include a gas cooler disposed adjacent to the evaporative cooling medium and configured to receive the first outflow of air and provide a second outflow of air. According to this example, a cooling system may include the cooling device, wherein the water header pipe may a first water header pipe and the evaporative cooling medium may be a first evaporative cooling medium. According to this example, the cooling system may further include a second water header pipe having a second plurality of drain holes arranged in a second row that is parallel with a second central axis of the second water header pipe, the second row may be oriented at a second angle measured between a second radius to the second central axis of the second water header pipe and a second vertical plane passing through the second central axis, the second water header pipe may be configured to receive the supply of water and discharge a sixth portion of water through the second plurality of drain holes.

[0009] According to this example, this system may include a second evaporative cooling medium that may include a third media portion and a fourth media portion, the third media portion may have a third media first end, a third media second end, and a third media body, the third media first end may be configured to receive the sixth portion of water, the third media body may be configured to conduct the received sixth portion of water within the third media body as a seventh portion of water, the third media second end may be configured to discharge an eighth portion of water; and the fourth media portion may have a fourth media first end, a fourth media second end, and a fourth media body, the fourth media first end may be configured to receive the eighth portion of water, the fourth media body may be configured to conduct the received eighth portion of water within the fourth media body as a ninth portion of water, wherein the fourth media body may be configured to receive a second inflow of air at a fourth temperature and discharge a third outflow of air at a fifth temperature that may be lower than the fourth temperature, the third outflow of air may be cooled by evaporative cooling of the ninth portion of water.

[0010] According to this example, the cooling system wherein the second angle may be between at least 10-degrees to not more than 40-degrees, wherein the sixth portion of water may be directed to a third media portion exterior edge, wherein the eighth portion of water may be configured to initially saturate a fourth media portion exterior surface, wherein at least one of the third media portion and the fourth media portion may include cellulose. The cooling system may further include a second elongated deflector shield disposed to at least partially surround a second upper portion of the second water header pipe, the second elongated deflector shield may be configured to at least one of protect the second upper portion of the second water header pipe and redirect water from the second water header pipe in a direction toward the third media first end, wherein the second elongated deflector shield may include a second lower portion with a second lower edge portion that may be laterally displaced from a fourth vertical plane that is coplanar with a second outer surface of the third media portion. The cooling system may further include a first gas cooler disposed adjacent to the first evaporative cooling medium and configured to receive the first outflow of air and provide a second outflow of air, a second gas cooler disposed adjacent to the second evaporative cooling medium and configured to receive the third outflow of air and provide a fourth outflow of air. The cooling system may further include at least one fan unit configured to receive the second outflow of air and the fourth outflow of air and expel a fifth outflow of air.

[0011] According to an example, a cooling method may include discharging a first portion of water from a water header pipe with a plurality of drain holes arranged in a row that is parallel with a central axis of the water header pipe, the row oriented at an angle measured between a radius to the central axis of the water header pipe and a vertical plane that passes through the central axis. The cooling method may include receiving the first portion of water at a first media first end of a first media portion. The cooling method may include conducting the first portion of water within a first media body of the first media portion as a second portion of water, the first media body may include a first plurality of channels arranged in a substantially vertical manner. The cooling method may include discharging the second portion of water as a third portion of water from a first media second end of the first media portion. The cooling method may include receiving the third portion of water at a second media first end of a second media portion. The cooling method may include conducting the third portion of water within a second media body as a fourth portion of water, the second media body may include a second plurality of channels arranged in a substantially horizontal manner. The cooling method may include receiving a first inflow of air at a first temperature into the second media body. The cooling method may include cooling the first inflow of air by evaporative cooling of the fourth portion of water in the second media body to provide a first outflow of air at a second temperature that is lower than the first temperature. The cooling method may include discharging the first outflow of air.

[0012] According to this example, the cooling method wherein the angle may be between at least 10-degrees to not more than 40-degrees. The cooling method wherein the first portion of water may be directed to a first media portion exterior edge, and wherein the fourth portion of water may be configured to initially saturate a second media portion exterior surface. The cooling method may further include discharging the fourth portion of water as a fifth portion of water from a second media second end of the second media portion. The cooling method may include collecting the fifth portion of water as a collected fifth portion of water. The cooling method may include pumping at least a portion of the collected fifth portion of water to provide a supply of water to the water header pipe.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] For a more complete understanding of the present disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawing figures, in which:

[0014] FIG. **1** illustrates a side plan view of a cooling system incorporating a first cooling device, according to an example;

[0015] FIG. **2** illustrates a front plan view of a water header pipe, according to an example;

[0016] FIG. **3** illustrates an enlarged side plan view of a portion of the cooling system of FIG. **1**, according to an example;

[0017] FIG. **4** illustrates a partial perspective view of a cooling system incorporating the cooling device of FIG. **1**, according to an example;

[0018] FIG. **5** illustrates a partial perspective view of an elongated deflector shield partially surrounding an upper portion of the water pipe of the colling system of FIG. **1**, according to an example;

[0019] FIG. **6** illustrates a side plan view of a cooling system incorporating a second cooling device, according to an example;

[0020] FIG. **7** illustrates an end view of a cooling system incorporating two cooling devices, according to an example;

[0021] FIG. **8** illustrates a side view of a dual-fan cooling system incorporating two cooling devices, according to an example;

[0022] FIG. **9** illustrates a side view of a **7**-fan cooling system incorporating two cooling devices, according to an example; and

[0023] FIG. **10** illustrates a method of operating a cooling system, according to an example.

DETAILED DESCRIPTION

[0024] As will be described more fully below, an improved adiabatic cooler may include a water header pipe having a discharge point of water from a plurality of drain holes at angle that may not be directly downward. Instead, the discharge point may be disposed at an angle and may be directed at least partly towards an outside edge an adjacent cooling pad, for example. Various advantages from this arrangement may include reducing the likelihood that dirt or other contaminants that may reside or collect in a supply of water or which may accumulate in a bottom portion of a water header pipe may block passage of water through the plurality of drain holes. Further, displacing the plurality of drain holes from a directly downward row position may provide an exterior surface of the water header pipe that may be easier to clean, be less likely to corrode, and be more long-lasting.

[0025] Hereinafter, a description will be given of respective examples of the present disclosure with reference to the drawings. Note that the disclosure is merely an example, and appropriate changes, which maintain the spirit thereof and are easily conceivable by those skilled in the relevant art, are naturally included in the scope of the present disclosure. Moreover, in some cases, in the present specification and the respective drawings, the same reference numerals are assigned to elements similar to those mentioned above regarding the already-discussed drawings, and a detailed description thereof is omitted as appropriate.

[0026] FIG. **1** illustrates a side plan view of a cooling system **100** incorporating a first cooling device **102**, according to an example. As described, cooling system **100** may include one or more cooling devices **102** which include two or more elements described in reference to cooling system **100**. Cooling system **100** and cooling device **102** may include a water header pipe **110** that may include a plurality of drain holes **112** arranged in a row **114** that is parallel with a central axis **116** of water header pipe **110**. Row **114** may be orientated at an angle **118** that is measured between a radius **120** to central axis **116** of water header pipe **110** and a vertical plane **122** that passes through central axis **116**. For convenience, the structures described in the drawing figures may be compared with a three-dimensional, orthogonal coordinate system with an X-axis **104**, a Y-axis **106**, and a Z-axis **108**. As illustrated, vertical plane **122** may correspond to a Y-Z plane (**106**, **108**). Water header pipe **110** may be configured to receive a supply of water **124** and discharge a first portion of water **128** through plurality of drain holes **112**. Angle **118** may be between at least 10-degrees to not more than 40-degrees. For a thicker cooling pad (e.g., X-axis **104** dimension) and water header pipe centered, angle **118** may be greater than 10-40 degrees. For a thinner cooling pad, angle **118** may be less than 10-40 degrees.

[0027] Cooling device **102** may include an evaporative cooling medium **140** comprising a first media portion **142** and a second media portion **144**. Suitable evaporative cooling media **140**, such as the one or more evaporative cooling pads comprising first media portion **142** and second media portion **144**, may be obtained from Munters Corporation of Amesbury, MA 01913, USA. Suitable evaporative cooling pads may include cellulose material, such as cellulose paper, cardboard, or other material that may derive from natural or manufactured fiber sources such as plants, trees, tree bark, and the like. First media portion **142** may have the shape of a filled, 3-dimensional “box” or rectangular prism held in a frame, with a first media first end **150** oriented as a top-most portion of first media portion **142** in FIG. **1**, a first media second end **152** oriented as a bottom-most portion

of first media portion **142** in FIG. **1**, and a first media body **154** disposed between first media first end **150** and first media second end **152**. First media first end **150** may be configured to receive first portion of water **128**. First media portion **142** may include a first plurality of channels **146** within first media portion **142** arranged in a substantially vertical manner, e.g., substantially along Y-axis **106**, as illustrated. As used herein, the term substantially may be used as similar terms effectively or practically. Hence, the term substantially along X-axis **104** or along Y-axis **106** may include the sense that the relative alignment of these elements may be the same as these reference directions in an effective or practical sense, without precise measuring, to allow for small variations while performing the same operations, for example. First media body **154** may be configured to receive and conduct first portion of water **128** within first media body **154** as a second portion of water **156**. First media second end **152** may be configured to discharge a third portion of water **158**.
[0028] Similarly, second media portion **144** may have the shape of a filled, **3**-dimensional “box” or rectangular prism held in a frame, with a second media first end **170**, a second media second end **172**, and a second media body **174** disposed between second media first end **170** and second media second end **172**. Second media first end **170** may be configured to receive and conduct third portion of water **158** within second media body **174** as a fourth portion of water **176**. Second media body **174** may include a second plurality of channels **148** within second media portion **144** arranged in a substantially horizontal manner, e.g., substantially along X-axis **104**, as illustrated. In this manner, fourth portion of water **176** may be distributed horizontally (e.g., laterally) within second media body **174** beginning with an exterior portion of second media body **174**, as will be described more fully below. Thus, second media body **174** may become saturated with water received from supply of water **124**, through first media body **154**, and discharged from first media body **154** to second media portion **144**. Second media body **174** may receive and conduct fourth portion of water **176** so that second media body **174** may receive an inflow of air **180** at a first temperature **182** (e.g., ambient temperature **182**) and discharge an outflow of air **184** at a second temperature **186** that is lower than the first temperature, where outflow of air **184** may be cooled by evaporative cooling of or by fourth portion of water **176**. In this manner, inflow of air **180** may pass through second media body **174** which contains water and emerge as outflow of air **184** traveling laterally along X-axis **104**, as illustrated. As used herein, the process of running water over or through an evaporative cooling pad and drawing air through the evaporative cooling pad may lower the ambient dry bulb temperature of the in-flowing air to provide adiabatic cooling of the out-flowing air from the pad, for example.
[0029] Some portion of inflow of air **180** may travel over first media first end **150** and may result in a cross-flow component of air flow across first media first end **150** in a direction from an outside surface of first media portion **142** to an inside surface of first media portion **142**, as will be described more fully below. Second media body **174** may be configured to distribute received fourth portion of water **176** and discharge a fifth portion of water **178** which may collect in a reservoir **192** (e.g., a tray) as a collected portion of water **194**. As described, third portion of water **158** may be introduced initially near an edge portion of second media portion **144**, fourth portion of water **176** may extend from the edge portion of second media portion **144** toward an opposite edge so that fifth portion of water **178** may be emitted from an entirety of second media second end **172**. A pump **196** may be operatively coupled with collected portion of water **194** (e.g., at least partially submerged) and configured to receive and transport collected portion of water **194** and provide supply of water **124** through a conduit, pipe, or other structure back to water header pipe **110**. Thus, collected portion of water **194** may provide a supply of water **124** that may flow through evaporative cooling medium **140** to cool inflow of air **180** by evaporative cooling and then be recycled. Additional water may be supplied to compensate for water lost due to evaporation.
[0030] According to an example, cooling system **100** and cooling device **102** may further include an elongated deflector shield **160** disposed to at least partially surround (e.g., placed at least partially around) an upper portion **162** of water header pipe **110**. Elongated deflector shield **160**

may be configured to at least one of protect upper portion **162** of water header pipe **110** from impact damage and contaminants, and an inside or underside portion of elongated deflector shield **160** may redirect water from water header pipe **110** in a direction toward first media first end **150**. Elongated deflector shield **160** may also include a lower portion **164** with a lower edge portion **166** that is laterally displaced from a second vertical plane **126** that is coplanar (e.g., a plane parallel with) with an outer surface **145** of first media portion **142** and parallel with vertical plane **122**. In this manner, some amount of first portion of water **128** that may contact an inside/underside surface of elongated deflector shield **160** or an inside surface of lower portion **164** may be redirected toward first media first end **150**. First portion of water **128** may be directed toward first media first end **150** by reflection off the inside surface of lower portion **164** or may fall under the force of gravity in a vertical manner (e.g., opposite Y-axis **106** direction, as illustrated) onto first media first end **150**.

[0031] As described above, the discharge point of water from water header pipe **110** from plurality of drain holes **112** at angle **118** may not be directly downward (e.g., opposite Y-axis **106** direction). Instead, first portion of water **128** may be discharged at angle **118** and may be directed at least partly towards an outside edge of first media portion **142**, as will be described more fully below. Various benefits and technical effects may accrue from this arrangement, where first portion of water **128** may be discharged at an angle, instead of directly downward. Such benefits of displacing plurality of drain holes **112** from a directly downward row position (e.g., a negative Y-axis **106** direction) to an angularly displaced row position as described may include reducing the likelihood that dirt or other contaminants that may reside or collect in supply of water **124** or which may accumulate in a bottom portion of water header pipe **110** may block passage of water through plurality of drain holes **112**. Some debris, minerals, or sediment may accumulate within water header pipe **110** and may be removed during periodic maintenance, but such accumulated sediment would not immediately block plurality of drain holes **112**, for example. Further, displacing plurality of drain holes **112** from a directly downward row position may provide an exterior surface of water header pipe **110** that may be easier to clean, be less likely to corrode, and be more long-lasting. Additionally, first portion of water **128** may be more likely to contact first media first end **150** if some amount of first portion of water **128** is displaced by a cross-flow component of air flowing across first media first end **150**. Stated differently, a higher proportion of discharged first portion of water **128** may be more likely to land on and be captured by first media first end **150** resulting in more water captured within first media portion **142** and subsequently conducted to second media portion **144**, and less water waste resulting in higher recycle efficiency.

[0032] Cooling system **100** may further include a gas cooler **130** disposed adjacent to the cooling device **102**, where gas cooler **130** may be configured to receive hot gas **132** (e.g., a heated coolant or refrigerant material such as carbon dioxide) at a gas cooler inlet **134** and provide cooled gas **136** at a gas cooler outlet **138**. Gas cooler **130** may include one or more refrigerant conducting pipes **131** (e.g., coils) which may be surrounded by heat radiating fins **133**, for example. Typically, a gas cooler is configured to reduce the temperature of a circulating coolant gas without changing the state of the coolant from gas to liquid. However, this is not considered limiting.

[0033] Separately, gas cooler **130** may be configured to use air to cool a circulating gas, while cooling device **102** may be configured to reduce a surrounding air temperature. Together, cooling device **102** and gas cooler **130** may comprise a gas cooling unit **139** in cooling system **100**, where cooling device **102** may reduce surrounding air temperature that is applied to gas cooler **130**. In this manner, cooling device **102** may be considered as a “pre-cooler” for gas cooler **130** and may efficiently improve the operation of gas cooler **130**. Second media body **174** may receive and conduct fourth portion of water **176** so that second media body **174** may receive inflow of air **180** at a first temperature **182** and discharge outflow of air **184** at a second temperature **186** that is lower than the first temperature, where outflow of air **184** may be cooled by evaporative cooling of fourth portion of water **176**. As mentioned above, the process of running water over or through an evaporative cooling pad such as second media portion **144** and drawing air through the evaporative cooling pad may lower the ambient dry bulb temperature of the in-flowing air to provide adiabatic cooling of the out-flowing air from the pad. Outflow of air **184** may be applied to gas cooler **130** which may discharge a second outflow of air **188** at a second outflow temperature **190**. Compared with a stand-alone gas cooler, the combination of cooling device **102** with gas cooler **130** provides a lower intermediate airflow temperature **186** compared with ambient temperature **182** without cooling device **102**.

[0034] FIG. **2** illustrates a front plan view of water header pipe **110**, according to an example. In addition to the description of FIG. **1**, FIG. **2** illustrates how water header pipe **110** may include plurality of drain holes **112** arranged in row **114** that is parallel with central axis **116** (e.g., center of cylinder) of water header pipe **110**. Further, water header pipe **110** may receive water from supply of water **124** into a center portion **202** of water header pipe **110**, a first end portion **204** of water header pipe **110**, or a second end portion **206** of water header pipe **110**. As described, center portion **202** may be any position between first end portion **204** and second end portion **206**. A coupling from supply of water **124** to water header pipe **110** may be made using a T-fitting, a sleeve fitting, or an angle fitting, for example. Alternatively, supply of water **124** may be coupled with a combination of components such as a pipe or other conduit from another water header pipe (not shown in this view). At least one end portion (**204**, **206**) of water header pipe **110** may be closed and/or capped to prevent outflow of water and to maintain pressure during expulsion of first portion of water **128** from water header pipe **110** from pump **196**, for example. Pump **196** may operate according to a pump profile (e.g., water volume and water pressure) that is limited to at least one of: avoid overspray, avoid reflecting or bouncing off (e.g., ricochet) first media first end **150**, to reduce water waste, and promote capture and conduction of water in the first media portion **142**.

[0035] FIG. **3** illustrates an enlarged side plan view of a portion of the cooling system of FIG. **1**, according to an example. In addition to the description of FIGS. **1-2**, FIG. **3** illustrates how first portion of water **128** may be directed indirectly or directly to a first media portion exterior edge **302** (e.g., a top-front corner) of first media portion **142**. In this manner, an exterior side portion **306** in the X-axis **104** direction of first media portion **142** may become saturated (e.g., filled with water from first portion of water **128**) first so that an outside portion of first media portion **142** may have a relatively higher water content (e.g., or a higher water density gradient) compared with other portions of first media portion **142**, for example. The substantially vertical orientation of first plurality of channels **146** within first media portion **142** may allow gravity to more quickly conduct first portion of water **128** through first media portion **142** to be discharged from first media second end **152** and thus enter second media first end **170** closer to a second media portion exterior surface **310** leading to second media body **174** being more reliably saturated with water on an outer surface exposed directly to air leading to more efficient cooling. First media portion **142** and/or second media portion **144** may include angled perforations that allow water to drain from the outer edges of the pads, effectively cleaning the external surface and removing accumulated dirt, for example. Further, angled perforations may limit some amount of water from entering the interior of the pads and which may limit water droplets being carried into gas cooler **130** and/or coils **131** via the airflow.

[0036] FIG. **4** illustrates a partial perspective view of a cooling system **400** incorporating the cooling device **102** of FIG. **1**, according to an example. In addition to the descriptions of FIGS. **1-3**, FIG. **4** illustrates cooling system **400** may include cooling system **102** with water header pipe **110** fed from supply of water **124** at a center portion **202** with a T-fitting, first media portion **142**, a second media portion **144**, and a gas cooler **130**. As described above, water header pipe **110** may provide first portion of water **128** through first media portion **142** and third portion of water **158** to second media portion **144**. In this example, cooling system **400** may include a fan portion **406** configured to expel air from cooling system **400** in a vertical (e.g., a Y-axis **106**) direction, thereby

pulling or drawing air through both a water-saturated second media portion **144** and gas cooler **130**.
[0037] FIG. **5** illustrates a partial perspective view of an elongated deflector shield **160** partially surrounding an upper portion **162** of water header pipe **110** of the cooling system of FIG. **1**, according to an example. In this example, water header pipe **110** may be fed by supply of water **124** at second end portion **206**, for example. Elongated deflector shield **160** may be mounted (e.g., suspended) above water header pipe **110** by suitable mounting hardware, for example. In various examples, water header pipe **110** and/or elongated deflector shield **160** may include various metal such as aluminum, steel, and stainless steel and/or plastic materials such as resin, a thermoplastic, or polyvinyl chloride (PVC).
[0038] FIG. **6** illustrates a side plan view of a cooling system **600** incorporating a second cooling device **602**, according to an example. In addition to the description of FIGS. **1-5**, FIG. **6** illustrates cooling system **600** may include one or more cooling devices **602** which include two or more elements described in reference to cooling system **600**. Similar to cooling system **100** and cooling device **102**, cooling system **600** and cooling device **602** may include a second water header pipe **610** that may include a second plurality of drain holes **612** arranged in a second row **614** that is parallel with a second central axis **616** of water header pipe **610**. As described, many elements of second cooling device **602** may be similar to or identical with corresponding elements of first cooling system **100** and/or first cooling device **102**, where at least some elements and features may be reflected about Y-axis **106**, as illustrated. Second row **614** may be orientated at a second angle **618** that may be measured between a second radius **620** to second central axis **616** of second water header pipe **610** and a second vertical plane **622** that is parallel to first vertical plane **122** and that passes through second central axis **616**. Second water header pipe **610** may be configured to receive supply of water **124** and discharge a sixth portion of water **628** through second plurality of drain holes **612**. Second angle **618** may be between at least 10-degrees to not more than 40-degrees.
[0039] Cooling device **602** may include an evaporative cooling medium **640** comprising a third media portion **642** and a fourth media portion **644**. Third media portion **642** may have the shape of a filled, 3-dimensional “box” or rectangular prism, with a third media first end **650** oriented as a top-most portion of third media portion **642** in FIG. **6**, a third media second end **652** oriented as a bottom-most portion of third media portion **642** in FIG. **6**, and a third media body **654** disposed between third media first end **650** and third media second end **652**. Third media first end **650** may be configured to receive sixth portion of water **628**. With brief reference to FIG. **3** and FIG. **6**, sixth portion of water **628** may be directed to a third media portion exterior edge (e.g., reference **302**). Third media body **642** may include a third plurality of channels **646** within third media portion **642** arranged in a substantially vertical manner, e.g., substantially along Y-axis **106**, as illustrated. Third media body **654** may be configured to receive and conduct sixth portion of water **628** within third media body **654** as a seventh portion of water **656**. Third media second end **652** may be configured to discharge an eighth portion of water **658**. As illustrated, second angle **618** may be arranged on an opposite side of (e.g., symmetrically reflected) second vertical plane **622** which is parallel with first vertical plane **122** as compared with first angle **118**. Thus, both first row **114** and second row **614** may be discharge first portion of water **128** and sixth portion of water **628**, respectively, outward (e.g., toward an outer edge) of second media portion **144** and fourth media portion **644**, respectively. With brief reference to FIG. **3** and FIG. **6**, eighth portion of water **658** may be configured to initially saturate a fourth media portion exterior surface (e.g., reference **310**).
[0040] Similarly, fourth media portion **644** may have the shape of a filled, **3**-dimensional “box” or rectangular prism, with a fourth media first end **670**, a fourth media second end **672**, and a fourth media body **674** disposed between fourth media first end **670** and fourth media second end **672**. Fourth media first end **670** may be configured to receive and conduct eighth portion of water **658** within fourth media body **674** as a ninth portion of water **676**. Fourth media body **674** may include a fourth plurality of channels **648** within fourth media portion **644** arranged in a substantially horizontal manner, e.g., substantially along X-axis **104**, as illustrated. In this manner, ninth portion

of water **676** may be distributed horizontally (e.g., laterally) within fourth media body **674** beginning with an exterior portion of fourth media body **674**, as will be described more fully below. Thus, fourth media body **674** may become saturated with water received from supply of water **124**, through third media body **654**, and discharged from third media body **654** to fourth media portion **644**. Fourth media body **674** may receive and conduct ninth portion of water **676** so that fourth media body **674** may receive a second inflow of air **680** at a fourth temperature **682** (e.g., ambient temperature **682**) and discharge a third outflow of air **684** at a fifth temperature **686** that may be lower than fourth temperature **682**, where third outflow air **684** may be cooled by evaporative cooling of ninth portion of water **676**. In this manner, second inflow of air **680** may pass through fourth media body **674** which contains water and emerge as third outflow of air **684** traveling laterally along X-axis **104** (e.g., in an opposite direction), as illustrated. Some portion of second inflow of air **680** may travel over third media first end **650** and may result in a cross-flow component of air flow across third media first end **650** in a direction from an outside surface of third media portion **642** to an inside surface of third media portion **642**. Fourth media body **674** may be configured to distribute received ninth portion of water **676** and discharge a tenth portion of water **678** which may collect in reservoir **192** as collected portion of water **194** along with discharged fifth portion of water **178**.

[0041] As described, eighth portion of water **658** may be introduced initially near an edge portion of fourth media portion **644**, ninth portion of water **676** may extend from the edge portion of fourth media portion **644** toward an opposite edge so that tenth portion of water **678** may be emitted from an entirety of fourth media second end **672**. As mentioned above, pump **196** may receive and transport collected portion of water **194** to provide supply of water **124**. In this manner, collected portion of water **194** may provide supply of water **124** that may flow through evaporative cooling medium **640** to cool second inflow of air **680** by evaporative cooling and then be recycled.

[0042] According to an example, cooling system **600** and second cooling device **602** may further include a second elongated deflector shield **660** disposed to at least partially surround (e.g., positioned at least partially around) a second upper portion **662** of second water header pipe **610**. Second elongated deflector shield **660** may be configured to at least one of protect second upper portion **662** of second water header pipe **610** from impact damage and contaminants, and an interior or underside portion of second elongated deflector shield **660** may redirect discharged water from second water header pipe **610** in a direction toward third media first end **650**. Second elongated deflector shield **660** may include a second lower portion **664** with a second lower edge portion **666** that is laterally displaced from a fourth vertical plane **626** that is parallel with a second outer surface **645** of third media portion **642** and parallel with second vertical plane **622**. In this manner, some amount of sixth portion of water **628** that may contact an inside or underside surface of second elongated deflector shield **660** or an inside or underside surface of second lower portion **664** may be redirected toward third media first end **650**. Sixth portion of water **628** may be directed toward third media first end **650** by reflection off the inside or underside surface of second lower portion **664** or may fall under the force of gravity in a vertical manner (e.g., opposite Y-axis **106** direction) onto third media first end **650**.

[0043] As described above, the discharge point of water from second water header pipe **610** from second plurality of drain holes **612** at second angle **618** may not be directly downward (e.g., opposite Y-axis **106** direction). Instead, sixth portion of water **628** may be discharged at second angle **618** and be directed at least partly towards an outside edge of third media portion **642**. Various benefits and technical effects may accrue from this arrangement, where sixth portion of water **628** may be discharged at an angle, instead of directly downward. As described above, such benefits of displacing second plurality of drain holes **612** from a directly downward row position to an angularly displaced row position as described may include reducing the likelihood that dirt or other contaminants that may reside or collect in supply of water **124** which may accumulate in a bottom portion of second water header pipe **610** may block passage of water through second plurality of drain holes **612**. Further, an exterior surface of water header pipe **110** may be easier to clean, be less likely to corrode, and be more long-lasting. Additionally, sixth portion of water **628** may be more likely to contact third media first end **650** if displaced by a cross-flow component of air flow across third media first end **650** resulting in higher recycle efficiency.

[0044] Cooling system **600** may further include a second gas cooler **630** disposed adjacent to second cooling device **602**, where second gas cooler **630** may be configured to receive hot gas **632** at a second gas cooler inlet **634** and provide cooled gas **636** at a second gas cooler outlet **638**. Second gas cooler **630** may include one or more refrigerant conducting pipes **631** surrounded by heat radiating fins **633**, for example. Separately, second gas cooler **630** may be configured to use air to cool a circulating gas, while cooling system **600** and second cooling device **602** may be configured to reduce a surrounding air temperature. Together, second cooling device **602** and second gas cooler **630** may comprise a second gas cooling unit **139**, where cooling device **602** may reduce surrounding air temperature that is applied to second gas cooler **630**. In this manner, second cooling device **602** may be considered as a “pre-cooler” for second gas cooler **630** and may efficiently improve the operation of second gas cooler **630**. As mentioned above, fourth media body **674** may receive and conduct ninth portion of water **676** so that fourth media body **674** may receive second inflow of air **680** at a fourth temperature **682** and discharge third outflow of air **684** at a fifth temperature **686** that is lower than fourth temperature **682**, where third outflow of air **684** may be cooled by evaporative cooling of ninth portion of water **676**. Third outflow of air **684** may be applied to second gas cooler **630** which may discharge a fourth outflow air **688** at a fifth outflow temperature **690**. Compared with a stand-alone gas cooler, the combination of second cooling device **602** with second gas cooler **630** provides a lower initial airflow temperature **686** compared with ambient temperature **682** without second cooling device **602**. Under various conditions, first temperature **182**, second temperature **186**, third temperature **190**, fourth temperature **682**, fifth temperature **686**, and sixth temperature **690** may be the same or different from each other.

[0045] FIG. **7** illustrates an end view of a cooling system **700** incorporating two cooling devices, according to an example. In addition to the descriptions of FIGS. **1-6**, FIG. **7** illustrates cooling system **700** that may include first cooling device **102** having first water header pipe **110** disposed on a first side of cooling system **700** (e.g., a right-side, as illustrated) to provide water to first evaporative cooling media **140** adjacent to first gas cooler **130**. Cooling system **700** may also include a second cooling device **602** that includes second water header pipe **610** disposed on a second side of cooling system **700** (e.g., a left-side as illustrated), opposite first cooling device **102**, to provide water to second evaporative cooling media **640** adjacent to second gas cooler **630**. In this manner, cooling system **700** may have two cooling devices (**102**, **602**) and two gas coolers (**130**, **630**) having similar or identical properties as these same elements described in reference to FIGS. **1-6**, for example. In this example, first evaporative cooling media **140** and second evaporative cooling media **640** may both span the width of first gas cooler **130** and second gas cooler **630** on their respective sides of cooling system **700**. First evaporative cooling media **140** may include multiple instances (e.g., panels) of first media portion **142** and multiple instances of second media portion **144** (shown in FIG. **1**), for example. Similarly, second evaporative cooling media **640** may include multiple instances (e.g., panels) of third media portion **642** and multiple instances of fourth media portion **644** (shown in FIG. **6**), for example.

[0046] First water header pipe **110** and second water header pipe **610** may be coupled together via a coupling pipe **711**, where either first water header pipe **110** or second water header pipe **610** may receive water directly from supply of water **124**, for example. Alternatively, both first water header pipe **110** and second water header pipe **610** may be sourced directly from supply of water **124**. In this example, first inflow of air **180** from a first direction at a first temperature **182** (e.g., ambient temperature **182**) may flow through at least a portion of first cooling device **102** and then through first gas cooler **130**. Similarly, second inflow air **680** from a second direction, which may be substantially opposite the first direction of first inflow of air **180**, may flow through at least a portion of second cooling system **602** and second gas cooler **630**. Other arrangements of cooling system **600** are possible, where the cooling devices (**102**, **602**) may be disposed on adjacent faces of cooling system **600**, for example. Cooling system **700** may also include a fan **750** configured to expel exhaust air **760** in a vertical (e.g., a Y-axis **106**) direction, for example. In this manner, first inflow of air **180** and second inflow of air **680** may be drawn from outside cooling system **700** into an inner cavity **706** and combined, then expelled as exhaust air **760**. Cooling system **700** may include one fan **750** or a plurality of fans, as will be described below.

[0047] FIG. **8** illustrates a side view of a dual-fan cooling system **800** incorporating two cooling devices, according to an example. In addition to the descriptions of FIGS. **1-7**, FIG. **8** illustrates a cooling system **800** that may include a first cooling device **102** having a first water header pipe **110** disposed on a first side of cooling system **800** (e.g., a foreground as illustrated) to provide water to first evaporative cooling media **140** adjacent to a first gas cooler **130**. Cooling system **800** may also include a second cooling device **602**, similar to first cooling device **102**, having second water header pipe **610** disposed on a second side of cooling system **800** (e.g., a background as illustrated), opposite first cooling device **102**, to provide water to second evaporative cooling media **640** adjacent to second gas cooler **630**. In this example, first evaporative cooling media **140** illustrated in the foreground of FIG. **8** and second evaporative cooling media **640** illustrated in the background of FIG. **8** may both span the width of first gas cooler **130** and second gas cooler **630** on their respective sides of cooling system **800**. First evaporative cooling media **140** may include multiple instances (e.g., panels) of first media portion **142** and multiple instances of second media portion **144** (shown in FIG. **1**), for example. Similarly, second evaporative cooling media **640** may include multiple instances (e.g., panels) of third media portion **642** and multiple instances of fourth media portion **644** (shown in FIG. **6**), for example. First water header pipe **110** and second water header pipe **610** may be coupled together via a coupling pipe **711**, where either first water header pipe **110** or second water header pipe **610** may receive water from supply of water **124** to provide water to evaporative cooling media **140** for both first cooling device **102** and second cooling device **602**, for example. Alternatively, both first water header pipe **110** and second water header pipe **610** may be sourced from supply of water **124** that is also coupled to coupling pipe **711** to provide water to evaporative cooling media **140** for both first cooling device **102** and second cooling device **602**. Cooling system **800** may also include first fan **750** and a second fan **751** configured to expel first exhaust air **760** and second exhaust air **761** in a vertical (e.g., a Y-axis **106**) direction, for example. In this manner, first inflow of air **180** and second inflow of air **680** may be drawn from outside cooling system **800** into an inner cavity **706** and combined, then ejected as first exhaust air **760** and second exhaust air **761**. Inner cavity **706** may be continuous or divided vertically between first fan **750** and second fan **751**. An advantage of inner cavity **706** being continuous is that air may be drawn through an entirety of first evaporative cooling media **140** and second evaporative cooling media **640** by first fan **750** if second fan **751** is inoperable, or vice versa.

[0048] FIG. **9** illustrates a side view of a **7**-fan cooling system **900** incorporating two cooling devices, according to an example. In addition to the descriptions of FIGS. **1-8**, FIG. **9** illustrates a cooling system **900** that may include a first cooling device **102** having a first water header pipe **110** disposed on a first side of cooling system **900** (e.g., a foreground as illustrated) to provide water to first evaporative cooling media **140** adjacent to a first gas cooler **130**. Cooling system **900** may also include a second cooling device **602**, similar to first cooling device **102**, having a second water header pipe **610** disposed on a second side of cooling system **900** (e.g., a background as illustrated), opposite first cooling device **102**, to provide water to second evaporative cooling media **640** adjacent to second gas cooler **630**. In this example, first evaporative cooling media **140** illustrated in the foreground of FIG. **9** and second evaporative cooling media **640** illustrated in the background of FIG. **9** both span the width of first gas cooler **130** and second gas cooler **630** on their respective sides of cooling system **900**. In this example, first evaporative cooling media **140** illustrated in the foreground of FIG. **9** and second evaporative cooling media **640** illustrated in the background of FIG. **9** may both span the width of first gas cooler **130** and second gas cooler **630** on their respective sides of cooling system **900**. First evaporative cooling media **140** may include multiple instances (e.g., panels) of first media portion **142** and multiple instances of second media portion **144** (shown in FIG. **1**), for example. Similarly, second evaporative cooling media **640** may include multiple instances (e.g., panels) of third media portion **642** and multiple instances of fourth media portion **644** (shown in FIG. **6**), for example. First water header pipe **110** and second water header pipe **610** may be coupled together via a coupling pipe **711**, where either first water header pipe **110** or second water header pipe **610** may receive water from supply of water **124** to provide water to evaporative cooling media **140** for both first cooling device **102** and second cooling device **602**, for example. Alternatively, both first water header pipe **110** and second water header pipe **610** may be sourced from supply of water **124** that is also coupled to coupling pipe **711** to provide water to evaporative cooling media **140** for both first cooling device **102** and second cooling device **602**. Cooling system **900** may also include a plurality of fans (e.g., first fan **750**, second fan **751**, . . . , seventh fan **756**) configured to expel exhaust air (e.g., first exhaust air **760**, second exhaust air **761**, . . . , seventh exhaust air **766**) in a vertical (e.g., a Y-axis **106**) direction, for example. In this manner, first inflow air and second inflow air may be drawn from outside cooling system **900** into an inner cavity **706** and combined, then expelled as exhaust air. Inner cavity **706** may be continuous or divided vertically between first fan **750** through seventh fan **756**.

[0049] FIG. **10** illustrates a method **1000** of operating a cooling system, according to an example. In addition to the descriptions of FIGS. **1-9**, FIG. **10** illustrates a method **1000** (e.g., a process) that may begin with a step of discharging **1006** first portion of water **128** from water header pipe **110** with plurality of drain holes **112** arranged in row **114** that may be parallel with central axis **116** of water header pipe **110**. Row **114** may be oriented at angle **118** measured between radius **120** to central axis **116** of water header pipe **110** and vertical plane **122** that passes through central axis **116**. Method **1000** may continue with a step of receiving **1010** first portion of water **128** at first media first end **150** of first media portion **142**. Method **1000** may continue with a step of conducting **1014** first portion of water **128** within first media body **154** of first media portion **142** as second portion of water **156**. First media body may include first plurality of channels **146** arranged in a substantially vertical manner **106** (e.g., a y-axis **106** direction, as illustrated). Method **1000** may continue with a step of discharging **1018** second portion of water **156** as third portion of water **158** from first media second end **152** of first media portion **142**. Method **1000** may continue with a step of receiving **1022** third portion of water **158** at second media first end **170** of second media portion **144**. Method **1000** may continue with a step of conducting **1026** third portion of water **158** within second media body **174** as fourth portion of water **176**. Second media body **174** may include second plurality of channels **148** arranged in a substantially horizontal manner **104** (e.g., an X-axis **104** direction, as illustrated).

[0050] Method **1000** may continue with a step of receiving **1030** first inflow of air **180** at a first temperature **182** into second media body **174**. Method **1000** may continue with a step of cooling **1034** the first inflow of air by evaporative cooling of (e.g., provided by) fourth portion of water **176** in second media body **174** to provide first outflow of air **184** at second temperature **186** that is lower than first temperature **182**, for example. Method **1000** may continue with a step of discharging **1038** first outflow of air **184**. Angle **118** may be between at least 10-degrees to not more than 40-degrees. First portion of water **128** may be directed to a first media portion exterior edge. Fourth portion of water **176** may be configured to initially saturate a second media portion exterior surface. Method **1000** may continue with a step of discharging **1042** fourth portion of water **176** as fifth portion of water **178** from second media second end **172** of second media portion **144**. Method **1000** may continue with a step of collecting **1046** fifth portion of water **178** as collected fifth portion of water **194**. Method **1000** may continue with a step of pumping **1052** at least a portion of collected fifth portion of water **194** to provide supply of water **124** to water header pipe **110**. Method **1000** may continue by returning to step **1006**, for example.

Simultaneously, a parallel method may continue with second water header pipe **610** and the structure described in reference to FIGS. **1-9**, in various examples.

[0051] Modifications, additions, or omissions may be made to processes of method **1000** depicted in FIG. **10** associated with FIGS. **1-9**, as described. Such processes may include more, fewer, or other operations. For example, various operations may be performed in parallel or in any suitable order, where such parallel operation or reordering is not prohibited by the above descriptions. Further, the cooling systems and methods described herein may also be performed as a method of exchanging heat. Finally, the described methods may be performed in conjunction with a computer readable medium including instructions that when executed cause a controller or computer to operate a cooling system, as described.

[0052] While several examples have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

[0053] In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

[0054] To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A cooling device, comprising: a water header pipe having a plurality of drain holes arranged in a row that is parallel with a central axis of the water header pipe, the row being oriented at an angle measured between a radius to the central axis of the water header pipe and a vertical plane passing through the central axis, the water header pipe being configured to receive a supply of water and discharge a first portion of water through the plurality of drain holes; and an evaporative cooling medium comprising a first media portion and a second media portion, the first media portion having a first media first end, a first media second end, and a first media body, the first media first end being configured to receive the first portion of water, the first media body being configured to conduct the received first portion of water within the first media body as a second portion of water, the first media second end being configured to discharge a third portion of water; and the second media portion having a second media first end, a second media second end, and a second media body, the second media first end being configured to receive the third portion of water, the second media body being configured to conduct the received third portion of water within the second media body as a fourth portion of water. wherein the second media body is configured to receive a first inflow of air at a first temperature and discharge a first outflow of air at a second temperature that is lower than the first temperature, the first outflow of air being cooled by evaporative cooling of the fourth portion of water.

2. The cooling device of claim 1, wherein the angle is between at least 10-degrees to not more than 40-degrees.

3. The cooling device of claim 1, wherein the first portion of water is directed to a first media portion exterior edge, and wherein the second portion of water is configured to initially saturate a second media portion exterior surface.
4. The cooling device of claim 1, wherein the first media body includes a first plurality of channels arranged in a substantially vertical manner, and wherein the second media body includes a second plurality of channels arranged in a substantially horizontal manner, orthogonal to the first plurality of channels.
5. The cooling device of claim 1, wherein at least one of the first media portion and the second media portion include cellulose.
6. The cooling device of claim 1, further comprising: an elongated deflector shield disposed to at least partially surround an upper portion of the water header pipe, the elongated deflector shield being configured to at least one of protect the upper portion of the water header pipe and redirect water from the water header pipe in a direction toward the first media first end.
7. The cooling device of claim 6, wherein the elongated deflector shield includes a lower portion with a lower edge portion that is laterally displaced from a second vertical plane that is coplanar with an outer surface of the first media portion.
8. The cooling device of claim 1, wherein the second media second end is configured to discharge a fifth portion of water, the device further comprising: a reservoir configured to collect the fifth portion of water as a collected fifth portion of water; and a pump configured to transport the collected fifth portion of water and provide the supply of water to the water header pipe.
9. The cooling device of claim 8, wherein the pump has a pump profile that is limited to at least one of avoid overspray, avoid bouncing off first media first end, and promote capture and conduction of water in the first media portion.
10. The cooling device of claim 8, wherein the supply of water is provided to one of a center portion of the water header pipe and an end portion of the water header pipe, wherein at least one end portion of the water header pipe is closed.
11. A cooling system, comprising the cooling device of claim 1, the system further comprising: a gas cooler disposed adjacent to the evaporative cooling medium and configured to receive the first outflow of air and provide a second outflow of air.
12. A cooling system comprising the cooling device of claim 1, wherein the water header pipe is a first water header pipe and the evaporative cooling medium is a first evaporative cooling medium, the system further comprising: a second water header pipe having a second plurality of drain holes arranged in a second row that is parallel with a second central axis of the second water header pipe, the second row being oriented at a second angle measured between a second radius to the second central axis of the second water header pipe and a second vertical plane passing through the second central axis, the second water header pipe being configured to receive the supply of water and discharge a sixth portion of water through the second plurality of drain holes; and a second evaporative cooling medium comprising a third media portion and a fourth media portion, the third media portion having a third media first end, a third media second end, and a third media body, the third media first end being configured to receive the sixth portion of water, the third media body being configured to conduct the received sixth portion of water within the third media body as a seventh portion of water, the third media second end being configured to discharge an eighth portion of water; and the fourth media portion having a fourth media first end, a fourth media second end, and a fourth media body, the fourth media first end being configured to receive the eighth portion of water, the fourth media body being configured to conduct the received eighth portion of water within the fourth media body as a ninth portion of water, wherein the fourth media body is configured to receive a second inflow of air at a fourth temperature and discharge a third outflow of air at a fifth temperature that is lower than the fourth temperature, the third outflow of air being cooled by evaporative cooling of the ninth portion of water.
13. The cooling system of claim 12, wherein the second angle is between at least 10-degrees to not

more than 40-degrees, wherein the sixth portion of water is directed to a third media portion exterior edge, wherein the eighth portion of water is configured to initially saturate a fourth media portion exterior surface, wherein at least one of the third media portion and the fourth media portion include cellulose.

14. The cooling system of claim 12, further comprising: a second elongated deflector shield disposed to at least partially surround a second upper portion of the second water header pipe, the second elongated deflector shield being configured to at least one of protect the second upper portion of the second water header pipe and redirect water from the second water header pipe in a direction toward the third media first end, wherein the second elongated deflector shield includes a second lower portion with a second lower edge portion that is laterally displaced from a fourth vertical plane that is coplanar with a second outer surface of the third media portion.

15. The cooling system of claim 12, further comprising: a first gas cooler disposed adjacent to the first evaporative cooling medium and configured to receive the first outflow of air and provide a second outflow of air. a second gas cooler disposed adjacent to the second evaporative cooling medium and configured to receive the third outflow of air and provide a fourth outflow of air.

16. The cooling system of claim 15, further comprising: at least one fan unit configured to receive the second outflow of air and the fourth outflow of air and expel a fifth outflow of air.

17. A cooling method, comprising: discharging a first portion of water from a water header pipe with a plurality of drain holes arranged in a row that is parallel with a central axis of the water header pipe, the row oriented at an angle measured between a radius to the central axis of the water header pipe and a vertical plane that passes through the central axis; receiving the first portion of water at a first media first end of a first media portion; conducting the first portion of water within a first media body of the first media portion as a second portion of water, the first media body includes a first plurality of channels arranged in a substantially vertical manner; discharging the second portion of water as a third portion of water from a first media second end of the first media portion; receiving the third portion of water at a second media first end of a second media portion; conducting the third portion of water within a second media body as a fourth portion of water, the second media body includes a second plurality of channels arranged in a substantially horizontal manner; receiving a first inflow of air at a first temperature into the second media body; cooling the first inflow of air by evaporative cooling of the fourth portion of water in the second media body to provide a first outflow of air at a second temperature that is lower than the first temperature; and discharging the first outflow of air.

18. The cooling method of claim 17, wherein the angle is between at least 10-degrees to not more than 40-degrees.

19. The cooling method of claim 17, wherein the first portion of water is directed to a first media portion exterior edge, and wherein the fourth portion of water is configured to initially saturate a second media portion exterior surface.

20. The cooling method of claim 17, further comprising: discharging the fourth portion of water as a fifth portion of water from a second media second end of the second media portion; collecting the fifth portion of water as a collected fifth portion of water; and pumping at least a portion of the collected fifth portion of water to provide a supply of water to the water header pipe.